



CSCE 670 - Information Storage and Retrieval

Week 3: More on Link Analysis

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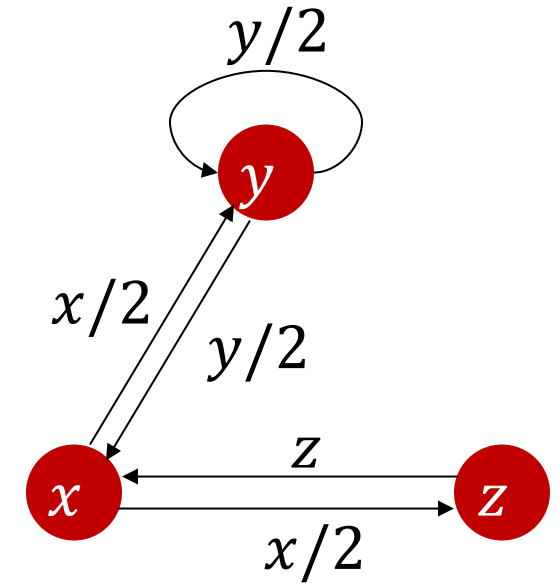
Course Website: <https://yuzhang-teaching.github.io/CSCE670-F26.html>

Recap: PageRank (Preliminary Version)

- Power Iteration:

- Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
- Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M}\mathbf{r}^{(t)}$
- Stop when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\| < \epsilon$

$$\mathbf{M} = \begin{bmatrix} 0 & 1/2 & 1 \\ 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 \end{bmatrix}$$



	$\mathbf{r}^{(0)}$	$\mathbf{r}^{(1)}$	$\mathbf{r}^{(2)}$	$\mathbf{r}^{(3)}$	...	Finally
x	1/3 (0.33)	1/2 (0.50)	1/3 (0.33)	11/24 (0.46)	...	0.40
y	1/3 (0.33)	1/3 (0.33)	5/12 (0.42)	3/8 (0.38)	...	0.40
z	1/3 (0.33)	1/6 (0.17)	1/4 (0.25)	1/6 (0.17)	...	0.20

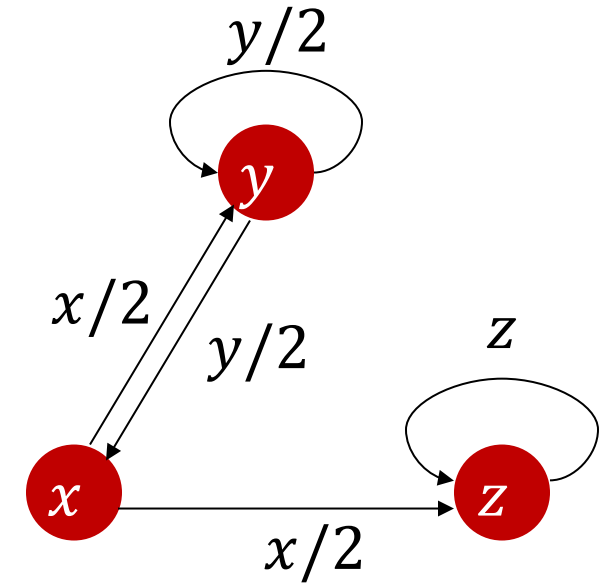
Can PageRank run into problems?

A Bad Example

- Power Iteration:

- Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
- Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M}\mathbf{r}^{(t)}$
- Stop when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\| < \epsilon$

$$\mathbf{M} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \end{bmatrix}$$



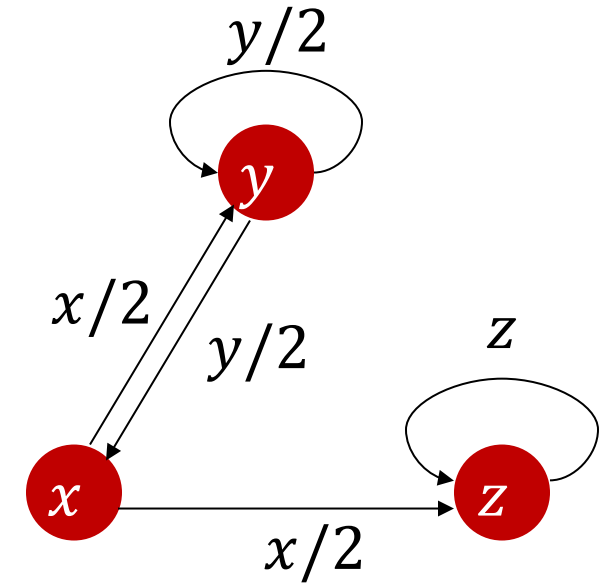
	$\mathbf{r}^{(0)}$
x	1/3 (0.33)
y	1/3 (0.33)
z	1/3 (0.33)

A Bad Example

- Power Iteration:

- Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
- Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M}\mathbf{r}^{(t)}$
- Stop when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\| < \epsilon$

$$\mathbf{M} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \end{bmatrix}$$



All the PageRank scores get “trapped” in node z.

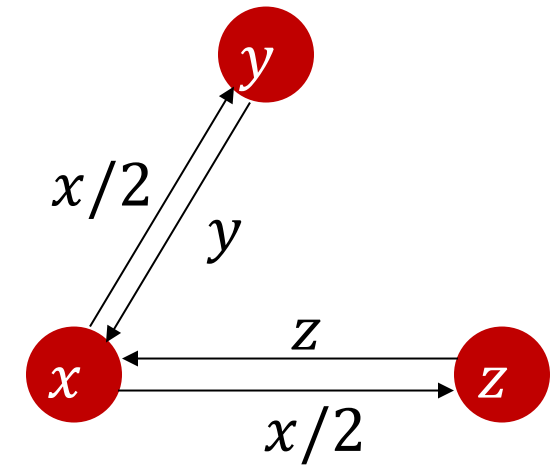
	$\mathbf{r}^{(0)}$	$\mathbf{r}^{(1)}$	$\mathbf{r}^{(2)}$	$\mathbf{r}^{(3)}$	...	Finally
x	1/3 (0.33)	1/6 (0.17)	1/6 (0.17)	1/8 (0.13)	...	0.00
y	1/3 (0.33)	1/3 (0.33)	1/4 (0.25)	5/24 (0.21)	...	0.00
z	1/3 (0.33)	1/2 (0.50)	7/12 (0.58)	2/3 (0.67)	...	1.00

An Even Worse Example

- Power Iteration:

- Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
- Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M}\mathbf{r}^{(t)}$
- Stop when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\| < \epsilon$

$$\mathbf{M} = \begin{bmatrix} 0 & 1 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 0 & 0 \end{bmatrix}$$



The algorithm falls into an infinite loop and will not terminate!
 Root cause: the graph is bipartite.

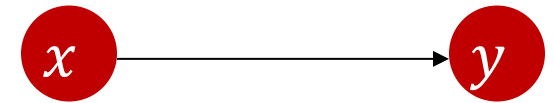
	$\mathbf{r}^{(0)}$	$\mathbf{r}^{(1)}$	$\mathbf{r}^{(2)}$	$\mathbf{r}^{(3)}$	...	Finally
x	1/3	2/3	1/3	2/3	...	?
y	1/3	1/6	1/3	1/6	...	?
z	1/3	1/6	1/3	1/6	...	?

Yet Another Even Worse Example

- Power Iteration:

- Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
- Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M}\mathbf{r}^{(t)}$
- Stop when $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\| < \epsilon$

$$\mathbf{M} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$



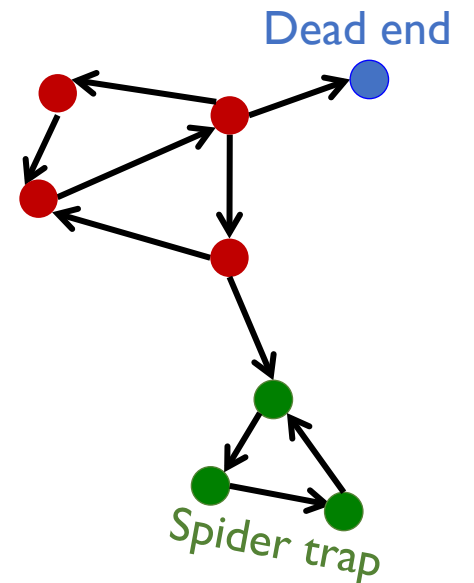
All the PageRank scores get “leaked”!

Root cause: the graph has a dead-end node (i.e., no out-links).

	$\mathbf{r}^{(0)}$	$\mathbf{r}^{(1)}$	$\mathbf{r}^{(2)}$	$\mathbf{r}^{(3)}$
x	1/2	0	0	0
y	1/2	1/2	0	0

Summary of the Challenges

- Spider traps
 - All out-links are within the group
 - Can have more than one node
 - Eventually spider traps absorb all importance
- Dead ends
 - The node has no out-links, therefore its importance score has nowhere to go
 - Eventually dead ends cause all importance to “leak out”
- Bipartite graph
 - If the graph is bipartite and the two partitions have different numbers of nodes, the algorithm will not converge.



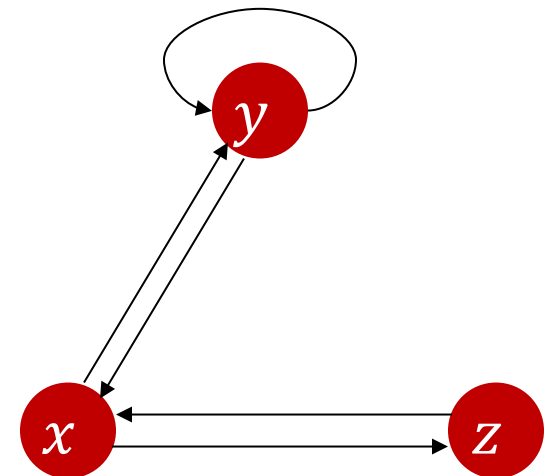
PageRank: Google Formulation

- Google's solution for spider traps: **Teleportation!**
 - Each node must contribute a portion of its importance score and distribute it evenly to all other nodes.

- Without teleports, $M = \begin{bmatrix} 0 & 1/2 & 1 \\ 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 \end{bmatrix}$

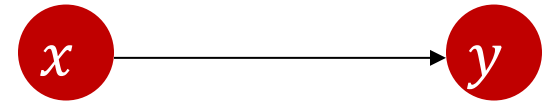
- With teleports, $M = \beta \begin{bmatrix} 0 & 1/2 & 1 \\ 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 \end{bmatrix} + (1 - \beta) \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$

- In practice, $\beta = 0.8, 0.85, \text{ or } 0.9$



How about dead ends?

- Dead ends must contribute **ALL** of its importance score and distribute it evenly to all other nodes.
- Without teleports, $M = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$
- Without teleports, $M = \beta \begin{bmatrix} 0 & 1/2 \\ 1 & 1/2 \end{bmatrix} + (1 - \beta) \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$
- Why do we call this solution “**teleportation**”?
 - Part of the importance score still flows according to the graph's defined neighborhoods
 - While the other part can instantly “**teleport**” to any node in the graph



Why does teleportation solve the problems?

- **Spider traps**: with traps, PageRank scores are not what we want
 - Solution: Never get stuck in a spider trap by teleporting out of it
- **Dead ends**: the matrix M is no longer column-stochastic (entries in a column may sum to 0 rather than 1)
 - Solution: Make M column-stochastic by always teleporting when there is nowhere else to go
- Wait, how about the **bipartite-graph issue**?
 - Teleportation makes the graph fully-connected (with different edge weights) and naturally non-bipartite.

PageRank: Google Formulation [Brin and Page, WWW 1998]

- Node-wise form:

$$r_j = \beta \left(\sum_{i \rightarrow j} \frac{r_i}{d_i} \right) + (1 - \beta) \frac{1}{N}$$

- **Note 1:** Each node i in the graph teleports a score of $(1 - \beta) \frac{1}{N} r_i$ to node j , so the total score node j receives through teleportation is exactly $(1 - \beta) \frac{1}{N} \sum_i r_i = (1 - \beta) \frac{1}{N}$.
- **Note 2:** This formulation assumes the graph has no dead ends. If there is a dead end, we can first link it to all the nodes (include itself).

PageRank: Google Formulation [Brin and Page, WWW 1998]

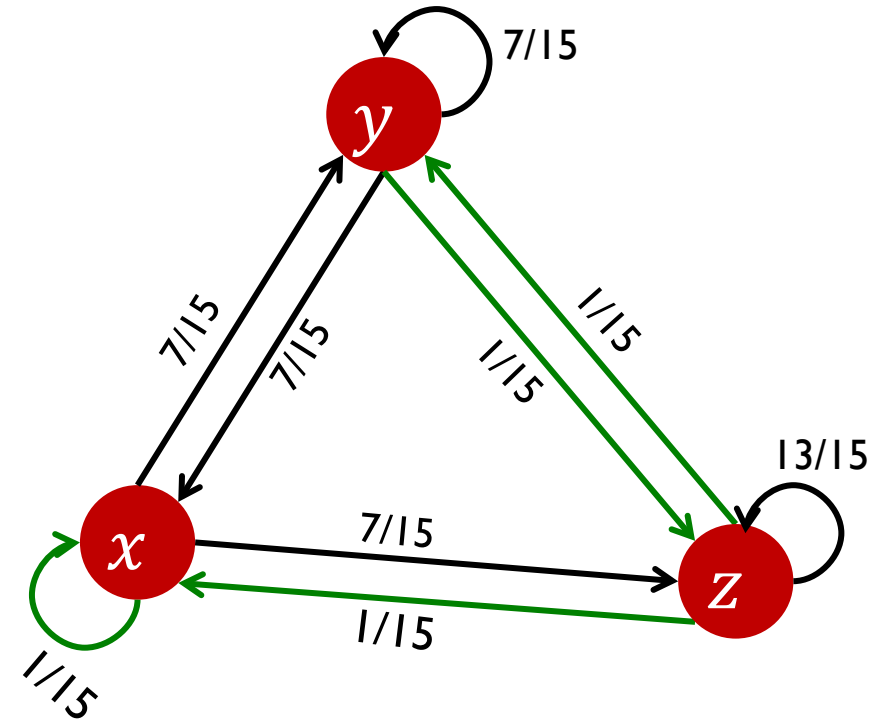
- Matrix form:

$$\mathbf{A} = \beta \mathbf{M} + (1 - \beta) \frac{\mathbf{1}}{N}$$

- **Note:** $\mathbf{1}$ is an $N \times N$ matrix where all entries are 1.
- Now we need to solve $\mathbf{A}\mathbf{r} = \mathbf{r}$
 - We can still use Power Iteration

Example ($\beta = 0.8$)

$$\begin{aligned}
 A &= 0.8 \times \begin{bmatrix} 0 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} + 0.2 \times \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \\
 &= \begin{bmatrix} 1/15 & 7/15 & 1/15 \\ 7/15 & 7/15 & 1/15 \\ 7/15 & 1/15 & 13/15 \end{bmatrix}
 \end{aligned}$$

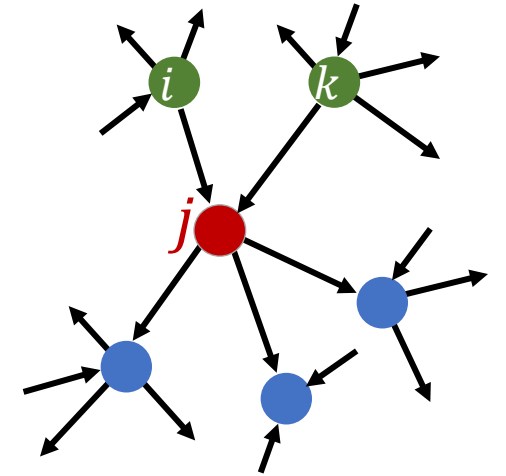


	$r^{(0)}$	$r^{(1)}$	$r^{(2)}$	$r^{(3)}$	...	Finally
x	1/3	0.20	0.20	0.18	...	0.15
y	1/3	0.33	0.28	0.26	...	0.21
z	1/3	0.47	0.52	0.56	...	0.64

Extended Content
(will not appear in quizzes or the exam)

PageRank: Random Walk Interpretation

- Imagine there is a random web surfer
 - At time t , the surfer is on a page i
 - At time $t + 1$, the surfer has two options
 - With probability β , it follows an out-link from i uniformly at random (i.e., ends up on some page j linked from i)
 - With probability $1 - \beta$, it jumps to a random page in the graph (can be i , j , or any other node)
- Let X_t denote the position of the surfer after t steps
- Let $p_j(t) = \text{Prob}[X_t = j]$ denote the probability that the surfer is on page j after t steps
 - $\sum_{j=1}^N p_j(t) = 1$
 - $\mathbf{p}(t) = [p_1(t), \dots, p_N(t)]$ is a **probability distribution**

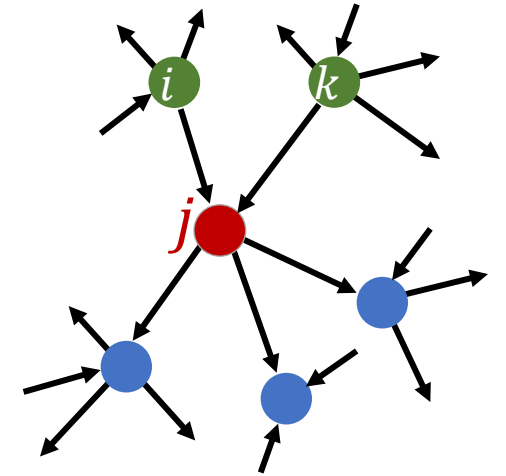


One Walk Step = One Matrix Multiplication

- The **transition probability** from page i to page j is
 - $A_{ji} = \text{Prob}[X_{t+1} = j \mid X_t = i]$
- To arrive at j , the surfer can come from any page i , so
 - $p_j(t + 1) = \text{Prob}[X_{t+1} = j]$
$$= \sum_{i=1}^N \text{Prob}[X_t = i] \cdot \text{Prob}[X_{t+1} = j \mid X_t = i]$$
$$= \sum_{i=1}^N p_i(t) \cdot A_{ji}$$
- In other words, $\mathbf{p}(t + 1) = \mathbf{A} \cdot \mathbf{p}(t)$
- Suppose the random walk reaches a state

$$\mathbf{p}(t) = \mathbf{p}(t + 1) = \mathbf{A} \cdot \mathbf{p}(t)$$

then $\mathbf{p}(t)$ is **stationary distribution** for the random walk



Why does (Power Iteration + Teleportation) work?

- **Contraction Mapping Theorem** (a.k.a., **Banach Fixed-Point Theorem**): Let F map probability distributions to probability distributions. If, for all distributions $\mathbf{p}$ and $\mathbf{q}$,
 - $\|F(\mathbf{p}) - F(\mathbf{q})\|_1 \leq c \cdot \|\mathbf{p} - \mathbf{q}\|_1$, where $0 < c < 1$ is a constant
- Then
 - A unique **fixed point** $\mathbf{r}$ exists: $F(\mathbf{r}) = \mathbf{r}$
 - From **any initial probability distribution** $\mathbf{x}$, repeated updates $F(\mathbf{x}), F(F(\mathbf{x})), \dots$ **converge to** $\mathbf{r}$
- Here, $\|\mathbf{p} - \mathbf{q}\|_1 = \sum_{i=1}^N |p_i - q_i|$
 - Measure the difference between two probability distributions

Teleportation makes PageRank a contraction

- Let $F(\mathbf{p}) = A\mathbf{p} = \beta M\mathbf{p} + (1 - \beta)\frac{1}{N}$
- Then $F(\mathbf{q}) = A\mathbf{q} = \beta M\mathbf{q} + (1 - \beta)\frac{1}{N}$
- So $\|F(\mathbf{p}) - F(\mathbf{q})\|_1 = \|\beta M\mathbf{p} - \beta M\mathbf{q}\|_1 = \beta\|M\mathbf{p} - M\mathbf{q}\|_1$
- When M is column-stochastic (dead ends fixed), it is easy to prove that $\|M\mathbf{p} - M\mathbf{q}\|_1 \leq \|\mathbf{p} - \mathbf{q}\|_1$
 - In other words, following links cannot increase L1 distance (**why?**)
- So $\|F(\mathbf{p}) - F(\mathbf{q})\|_1 \leq \beta\|\mathbf{p} - \mathbf{q}\|_1$
- Therefore, we can use the **Contraction Mapping Theorem**
 - A unique stationary distribution exists
 - Power iteration converges from any initial probability distribution

Back to the Broader Story of Ranking

Boolean + PageRank results for the query “university” [Page et al., 1999]

- With the rise of the Web, traditional **text-based signals** (e.g., TF-IDF and BM25) may not be sufficient.
- Many early web search engines relied on classic **text-based ranking** plus some rudimentary **link-based signals**.

The screenshot shows a web search engine interface with the query "university". The search bar at the top right contains "university" and a "Search" button. Below the search bar, there are tabs for "10 results", "clustering on", and "Search". The main results area displays a list of search results, each with a title, URL, and a score. The top result is "Optical Physics at the University of Oregon" with a score of 74.79%. Other results include "Stanford University Homepage", "Stanford University: Portfolio Collection", "University of Illinois at Urbana-Champaign", "Indiana University", "University of California, Irvine", "University of Minnesota", "Iowa State University Homepage", "The University of Michigan", "Mississippi State University", and "Northwestern University: NUInfo".

Query: university
11 Results Returned
Showing Results From 0 to 10

Stanford University Homepage
http://www.stanford.edu/
74.79% 4k - 2591993 - 010397

Stanford University: Portfolio Collection
http://www.stanford.edu/home/administration/portfolio.html
65.78% 3k - 2591993 - 010397

University of Illinois at Urbana-Champaign
http://www.uiuc.edu/
73.26% 13k - 1290096 - 010397

Indiana University
http://www.indiana.edu/
68.38% 1k - 092096 - 010597

University of California, Irvine
http://www.uci.edu/
68.07% 3k - 1290096 - 010397

University of Minnesota
http://www.umn.edu/
67.05% 0k - 1291696 - 010397

Iowa State University Homepage
http://www.iastate.edu/
66.66% 3k - 1291896 - 010397

The University of Michigan
http://www.umich.edu/
66.35% 1k - 2591993 - 010397

Mississippi State University
http://www.msstate.edu/
66.35% 3k - 2591993 - 010397

Northwestern University: NUInfo
http://www.nyu.edu/
66.15% 3k - 1291496 - 010597

Optical Physics at the University of Oregon
Oregon Center for Optics in Science and Technology. Department of Physics, University of Oregon, Eugene OR 97403. Research Groups: Carmichael Group....
http://optics.uoregon.edu/ - size 1K - 16 Dec 96

Carnegie Mellon University - Campus Networking
Departments. Data Communications. Data Communications is responsible for installing and maintaining all on campus networking equipment and all of...
http://www.net.cmu.edu/ - size 4K - 19 Aug 95

Wesleyan University Computer Science Group Home Page
Computer Science Group. Wesleyan University. Welcome to the home page of the Computer Science Group at Wesleyan University. We are administratively within.
http://www.cs.wesleyan.edu/ - size 2K - 15 Apr 96

Keio University Shonan Fujisawa Campus (SFC)
B\$3\$N%Z!EFnF#Bt%-&c%Q%9 (B(SFC) \$B\$N (BWWW \$B% \$BcmOU=q\$- (B \$B\$FI\$G\$/\$@5\$!\$ (B. Nihongo | English. SFC \$B>pJs (B. [\$B%a%G%#%*%?%?!*...
http://www.sfc.keio.ac.jp/ - size 3K - 5 Feb 97

School of Chemistry, University of Sydney
The School of Chemistry. School of Chemistry, University of Sydney, NSW 2006 Australia International Phone: +61-2-9351-4504 Fax: +61-2-9351-3329 Australia.
http://www.chem.su.oz.au/ - size 4K - 25 Feb 97

Mankato State University
The Campus Athletics, Campus Tour, Bookstore, Maps, Current Events... Admission & Registration Admissions, Financial Aid, Registrar's, Graduate...
http://www.mankato.msut.edu/ - size 3K - 27 Nov 96

St. Ambrose University
Main Index: Academic Departments. Administrative Services. Campus News. Computing Services. Galvin Fine Arts Center. Internet Connections. Library...
http://www.sau.edu/ - size 2K - 4 Feb 97

University of Washington ECSEL Projects

Back to the Broader Story of Ranking

- In practice, we will build a scoring function that considers many features.
- Typically, we have:
 - **Query-dependent features**: e.g., TF-IDF, BM25, # of times a query word occurs in a document, ...
 - **Query-independent features**: e.g., PageRank, # of in-links to a webpage, popularity of an album, ...
 - Many query-independent features are proxies for “reputation”
- **How to jointly consider these features?**
 - Week 5

Our Plan: Ranking

-  Why is ranking important?
-  What factors impact ranking?
- Two foundational text-based approaches
 -  TF-IDF
 -  BM25
- Two foundational link-based approaches
 -  PageRank
 - HITS
- Machine-learned ranking (“learning to rank”)

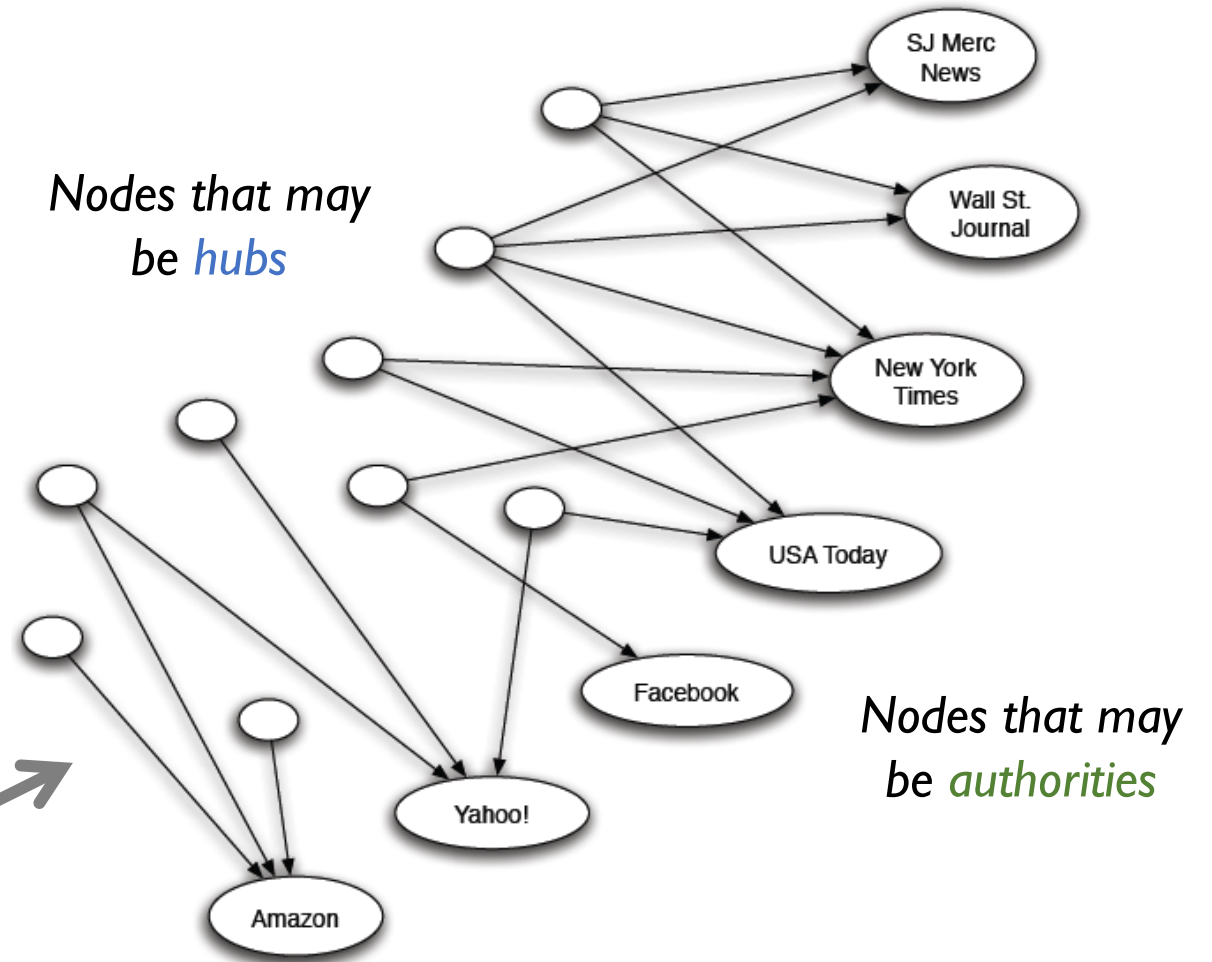
HITS

- **HITS (Hypertext-Induced Topic Selection)** [Kleinberg, SODA'98]
 - Is a measure of webpage importance, similar to PageRank
 - Proposed at around same time as PageRank
- **Goal:** Say we want to find good newspapers
 - Don't just find newspapers.
 - Find “experts” – people who link in a coordinated way to good newspapers
- **Idea:** Links as votes
 - Page is more important if it has more links
 - In-coming links? Out-going links?

Finding Newspapers

- Each page has 2 scores
 - Quality as content (**authority**)
 - Quality as an expert (**hub**)
- Interesting pages fall into two classes:
 - **Authorities** are pages containing useful information
 - **Hubs** are pages that link to authorities

Note this is idealized example. In practice, the graph is not bipartite, and each page has both **hub** and **authority** scores.

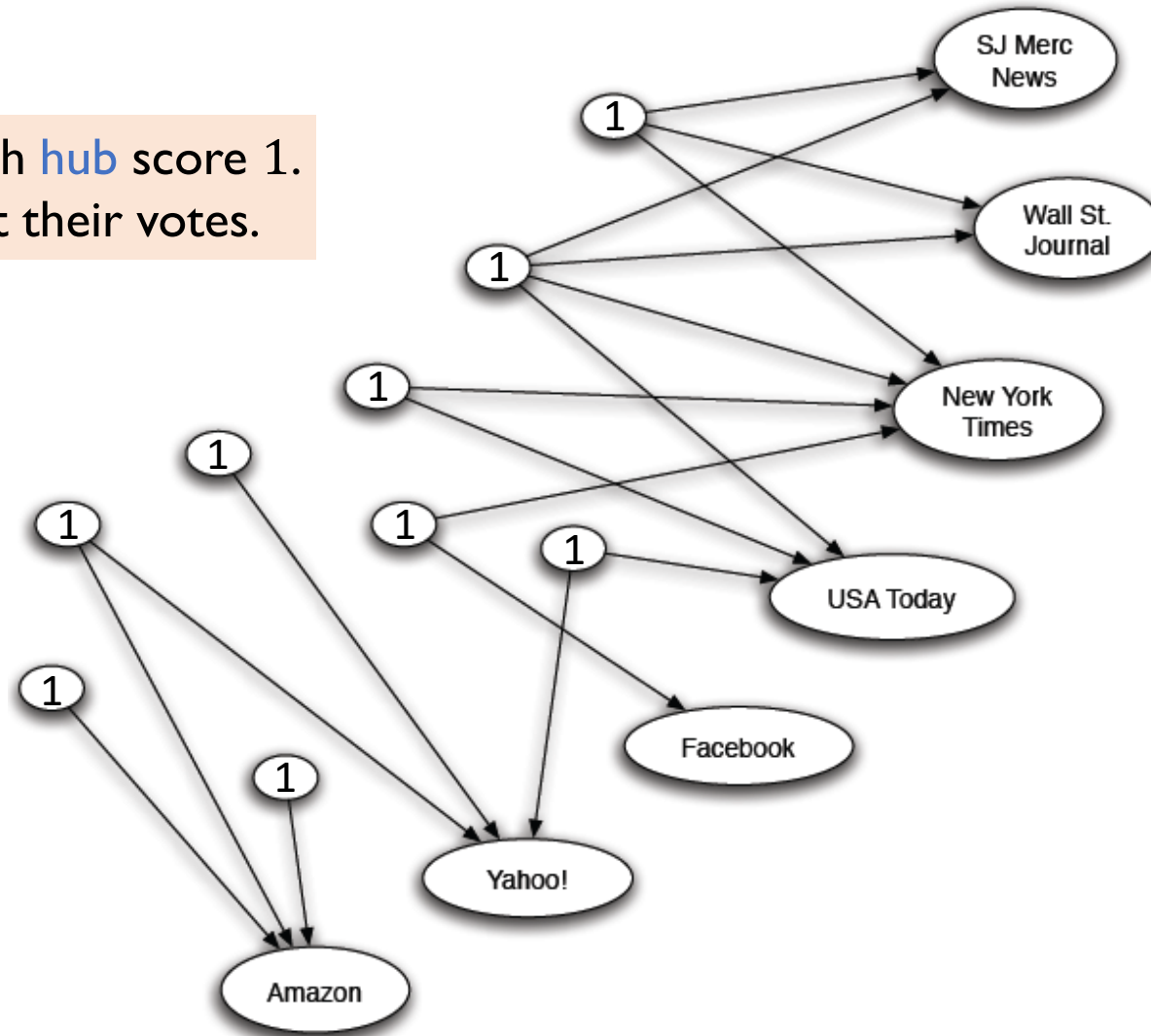


Hubs and Authorities

- **Authorities** are pages containing useful information
 - Newspaper homepages
 - Course homepages
 - Homepages of auto manufacturers
- **Hubs** are pages that link to authorities
 - List of newspapers
 - Course bulletin
 - List of US auto manufacturers
- **Mutually recursive** definition
 - A good **hub** links to many good **authorities**
 - A good **authority** is linked from many good **hubs**

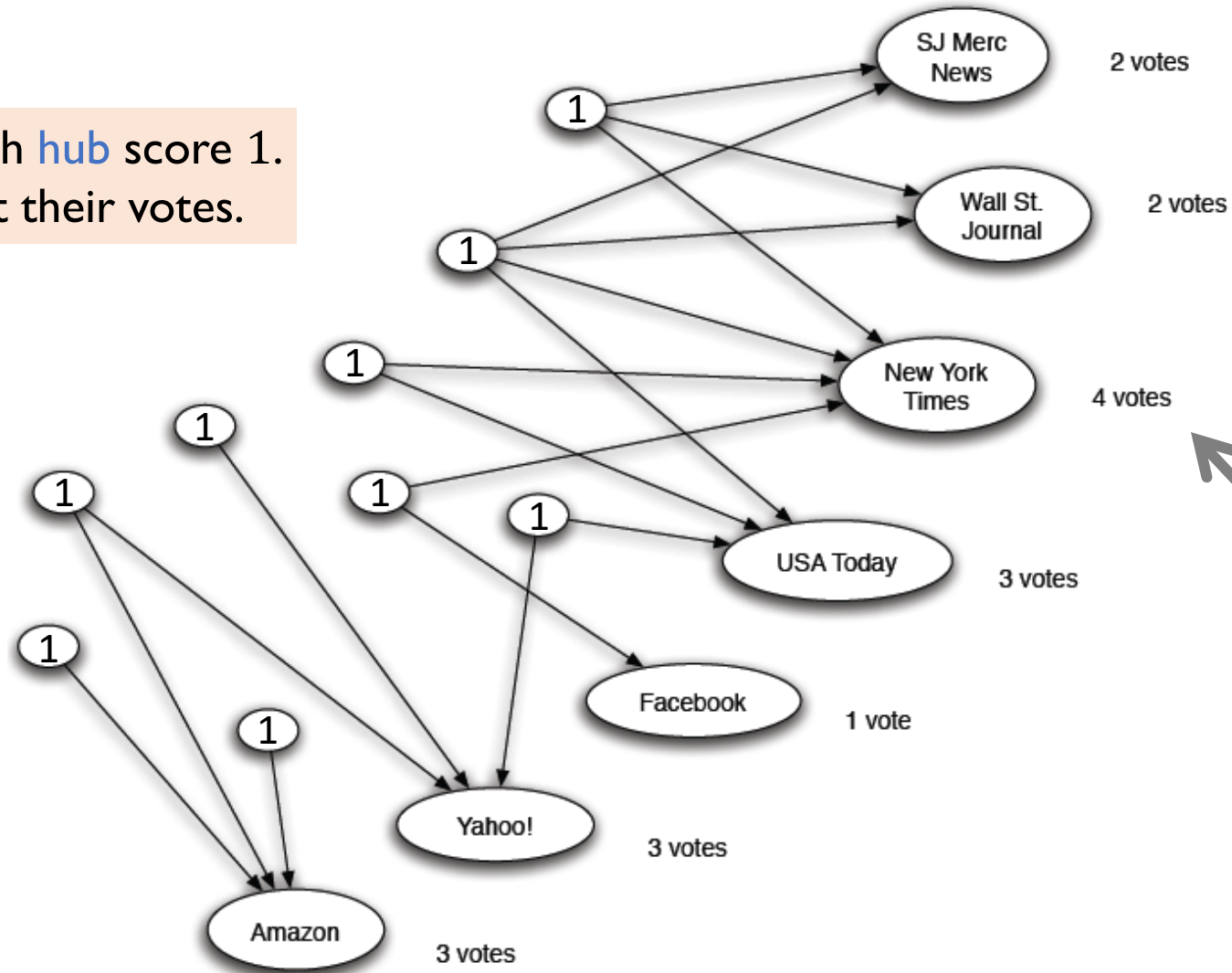
Principle of Repeated Improvement

Each page starts with **hub** score 1.
Authorities collect their votes.



Principle of Repeated Improvement

Each page starts with **hub** score 1.
Authorities collect their votes.

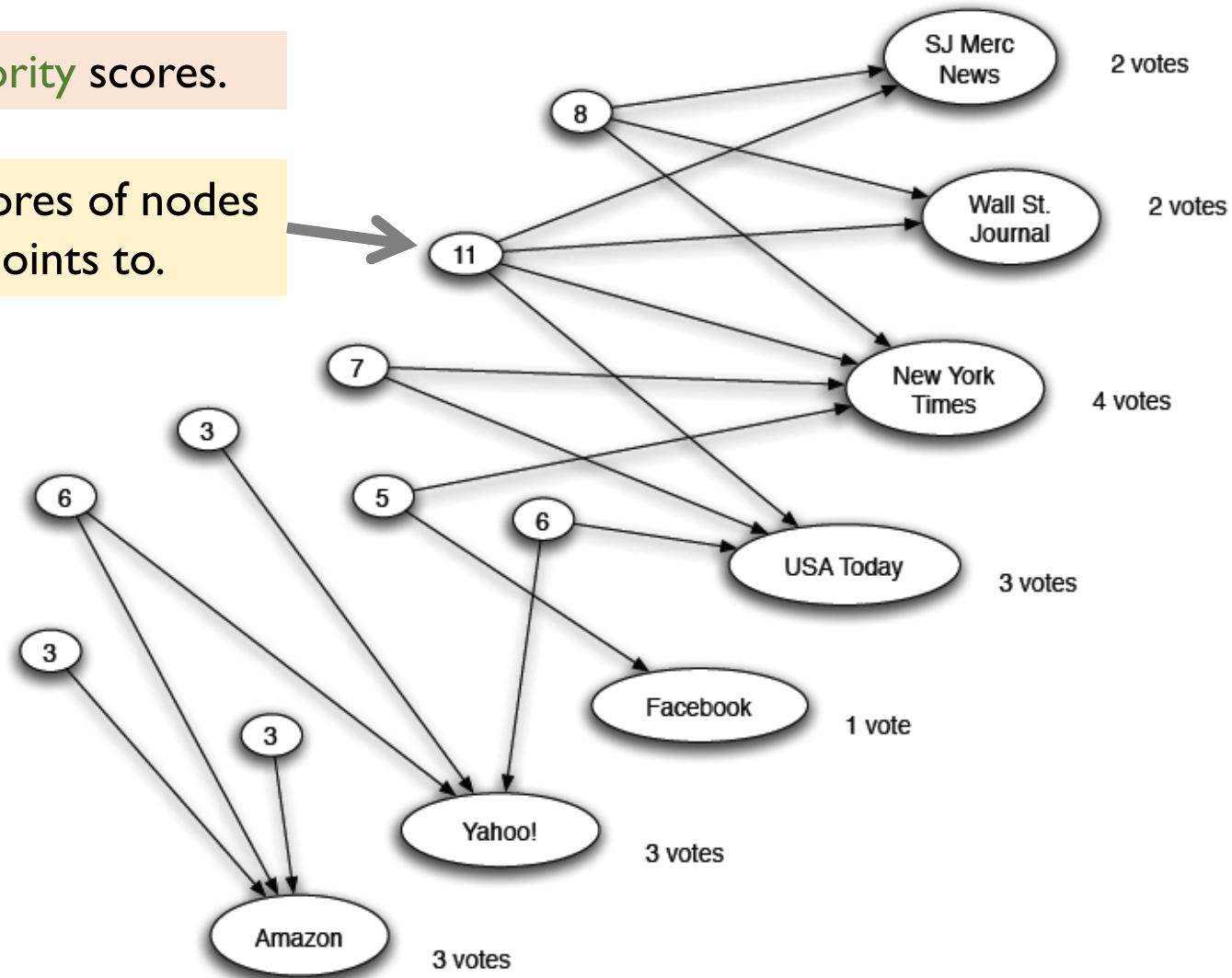


Sum of **hub** scores of nodes pointing to NYT

Principle of Repeated Improvement

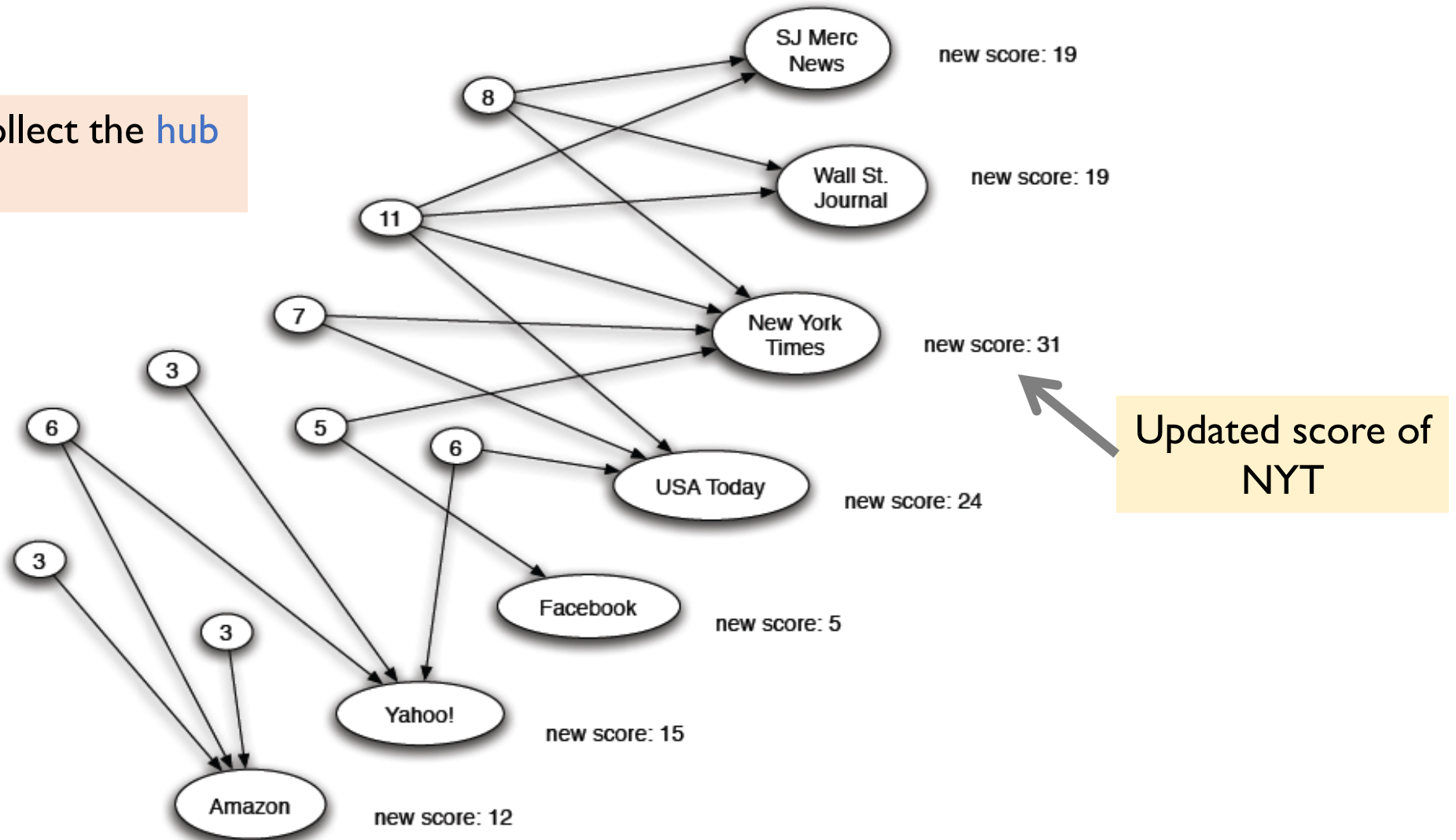
Hubs collect **authority** scores.

Sum of **authority** scores of nodes that the node points to.



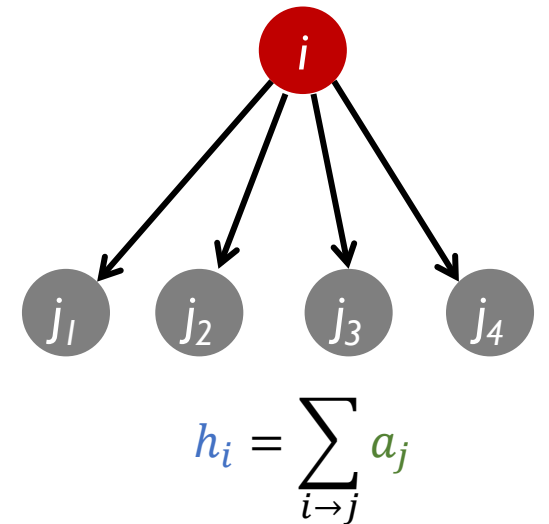
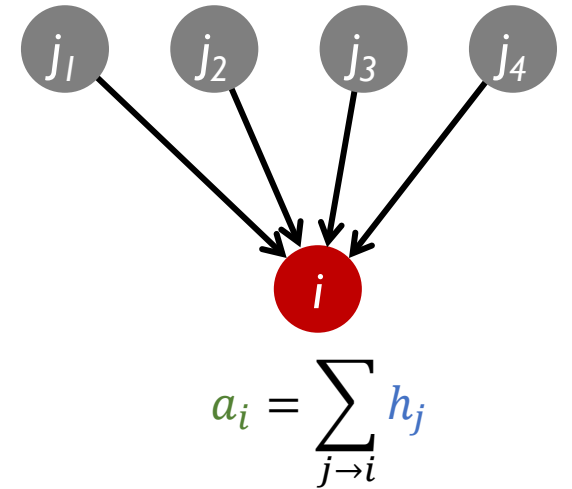
Principle of Repeated Improvement

Authorities again collect the hub scores.



HITS Algorithm: Formal Description

- Each page i has 2 scores:
 - Authority score: a_i
 - Hub score: h_i
- HITS algorithm
 - Initialize: $a_j^{(0)} = 1/\sqrt{N}$, $h_j^{(0)} = 1/\sqrt{N}$
 - Then keep iterating until convergence:
 - $\forall i$, update the authority score: $a_i^{(t+1)} = \sum_{j \rightarrow i} h_j^{(t)}$
 - $\forall i$, update the hub score: $h_i^{(t+1)} = \sum_{i \rightarrow j} a_j^{(t)}$
 - $\forall i$, normalize: $\sum_i \left(a_i^{(t+1)}\right)^2 = 1, \sum_j \left(h_j^{(t+1)}\right)^2 = 1$



Matrix Version

- Notation:

- Vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ \cdots \\ a_n \end{pmatrix}$ and $\mathbf{h} = \begin{pmatrix} h_1 \\ \cdots \\ h_n \end{pmatrix}$ denote the authority/hub scores of all pages

- Adjacency matrix $\mathbf{M}$, where $M_{ij} = \begin{cases} 1, & \text{if } i \rightarrow j \\ 0, & \text{otherwise} \end{cases}$

- Then, $h_i = \sum_{i \rightarrow j} a_j$ can be rewritten as $h_i = \sum_j M_{ij} a_j$

- In other words, $\mathbf{h} = \mathbf{M}\mathbf{a}$

- Similarly, $a_i = \sum_{j \rightarrow i} h_j$ can be rewritten as $a_i = \sum_j M_{ji} h_j$

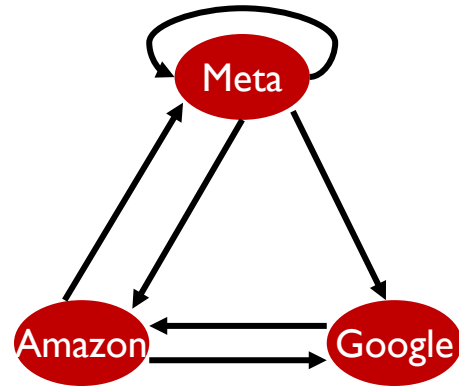
- In other words, $\mathbf{a} = \mathbf{M}^T \mathbf{h}$

Matrix Version

- $\mathbf{h} = \mathbf{M}\mathbf{a}$
- $\mathbf{a} = \mathbf{M}^T \mathbf{h}$
- If we ignore the normalization step
 - $\mathbf{a} = \mathbf{M}^T \mathbf{h} = \mathbf{M}^T \mathbf{M} \mathbf{a}$
 - Power Iteration with the matrix $\mathbf{M}^T \mathbf{M}$
 - $\mathbf{h} = \mathbf{M} \mathbf{a} = \mathbf{M} \mathbf{M}^T \mathbf{h}$
 - Power Iteration with the matrix $\mathbf{M} \mathbf{M}^T$
- Given the adjacency matrix $\mathbf{M}$,
 - The authority vector $\mathbf{a}$ we are looking for is an eigenvector of $\mathbf{M}^T \mathbf{M}$
 - The hub vector $\mathbf{h}$ we are looking for is an eigenvector of $\mathbf{M} \mathbf{M}^T$

Recall Power Iteration
in PageRank

Example



Meta Amazon Google

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Hub	$h^{(0)}$	$h^{(1)}$	$h^{(2)}$	$h^{(3)}$	...	Finally
Meta	0.58	0.80	0.80	0.79	...	0.788
Amazon	0.58	0.53	0.53	0.57	...	0.577
Google	0.58	0.27	0.27	0.23	...	0.211

Authority	$a^{(0)}$	$a^{(1)}$	$a^{(2)}$	$a^{(3)}$	...	Finally
Meta	0.58	0.58	0.62	0.62	...	0.628
Amazon	0.58	0.58	0.49	0.49	...	0.459
Google	0.58	0.58	0.62	0.62	...	0.628

PageRank and HITS

- PageRank and HITS are two solutions to the same problem:
 - How to identify important pages given the hyperlink graph of webpages?
- The destinies of PageRank and HITS after 1998 were very different



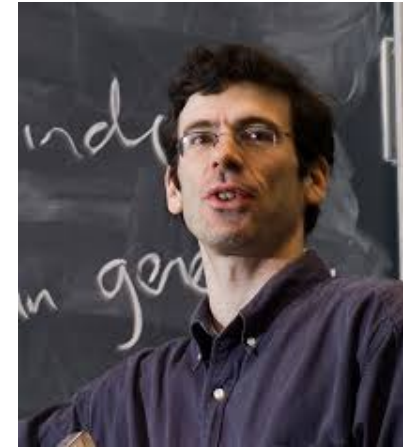
Sergey Brin



Larry Page

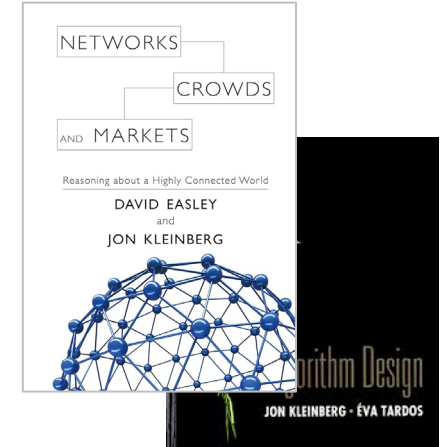


Co-founders of Google



Jon Kleinberg

Professor at Cornell University
Member of NAS and NAE



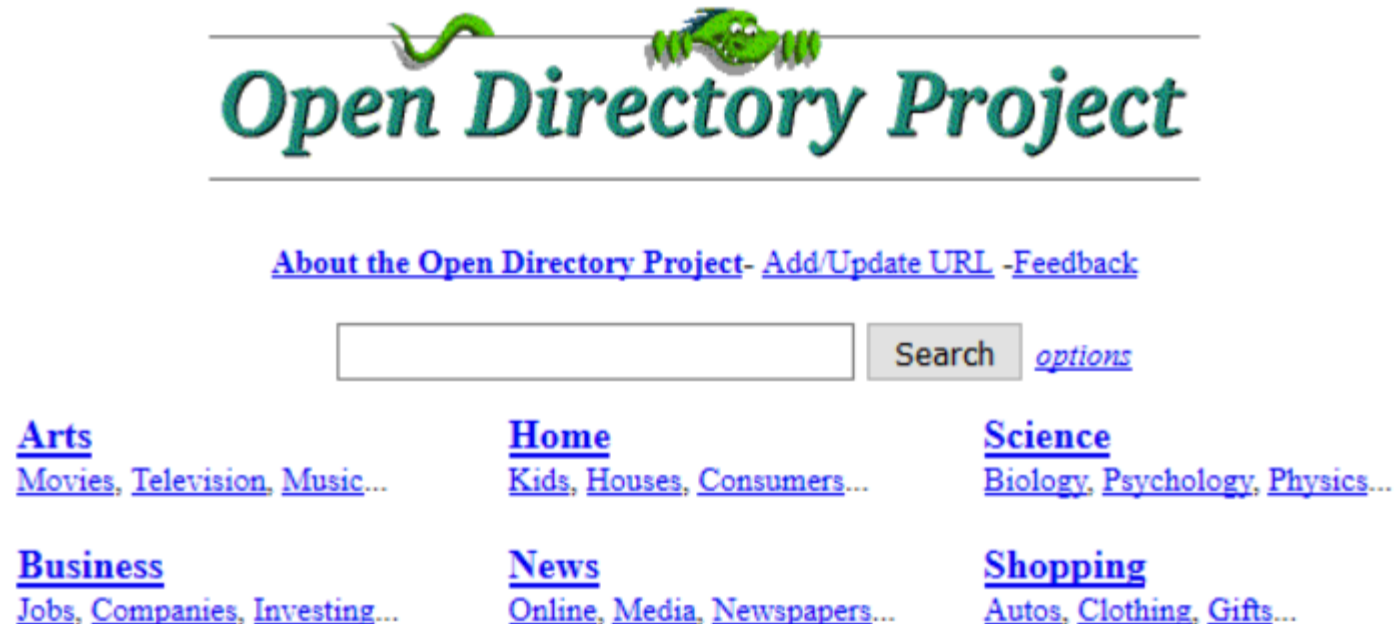
Topic-Sensitive PageRank

Topic-Sensitive PageRank (a.k.a., Personalized PageRank)

- PageRank measures **generic** importance of a page
 - Can we measure page importance **within a topic**?
- **Goal**: Evaluate Web pages not just according to their popularity, but by how close they are to a particular topic, e.g., “*sports*” or “*history*”
 - Allow search queries to be answered based on interests of the user
- **Idea**: Modify the teleportation mechanism
 - **Standard PageRank**: The random surfer can **teleport to any page** with equal probability
 - To avoid dead-end and spider-trap problems
 - **Topic-Sensitive PageRank**: The random surfer can only **teleport to a topic-specific set of “relevant” pages**

Topic-Sensitive PageRank (a.k.a., Personalized PageRank)

- **Topic-Sensitive PageRank**: The random surfer can only teleport to a topic-specific set of “relevant” pages (denoted as S)
 - S contains only pages that are relevant to the topic
 - E.g., Open Directory (DMOZ) pages for a given topic/query



Matrix Formulation

- Standard PageRank

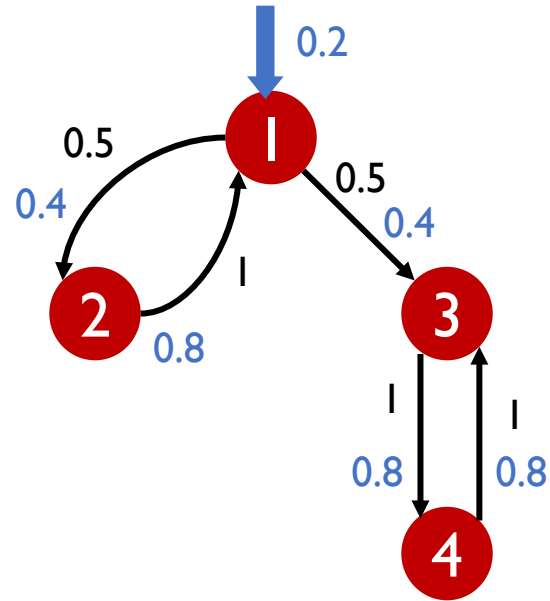
$$A_{ij} = \beta M_{ij} + (1 - \beta) \frac{1}{N}, \quad \forall \text{ pages } i, j$$

- Topic-Sensitive PageRank

$$A_{ij} = \begin{cases} \beta M_{ij} + (1 - \beta) \frac{1}{|S|}, & \text{if } i \in S \\ \beta M_{ij}, & \text{otherwise} \end{cases}$$

- We weighted all pages in S equally
 - Could also assign different weights to pages!
- The computation is similar to that of standard PageRank
 - Power Iteration

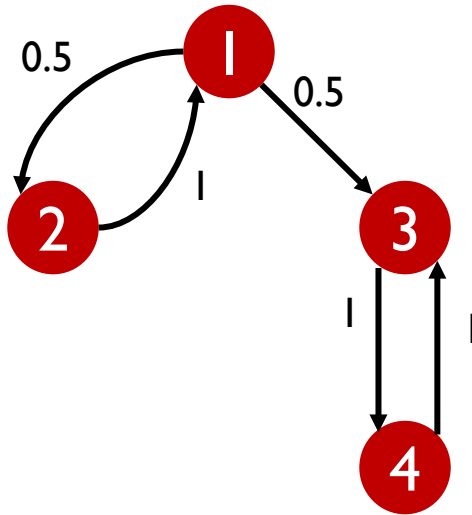
Example



Suppose $S = \{1\}$ and $\beta = 0.8$

	$r^{(0)}$	$r^{(1)}$	$r^{(2)}$	...	Finally
1	0.25	0.40	0.28	...	0.294
2	0.25	0.10	0.16	...	0.118
3	0.25	0.30	0.32	...	0.327
4	0.25	0.20	0.24	...	0.261

Example



$$S = \{1\}$$
$$\beta = 0.9$$

Node	Score
1	0.17
2	0.07
3	0.40
4	0.36

$$S = \{1\}$$
$$\beta = 0.8$$

Node	Score
1	0.29
2	0.12
3	0.33
4	0.26

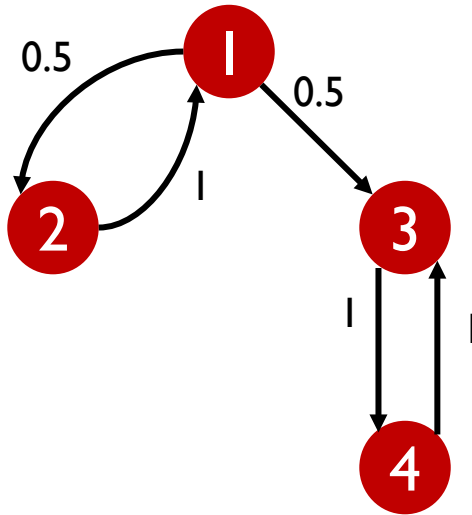
$$S = \{1\}$$
$$\beta = 0.7$$

Node	Score
1	0.39
2	0.14
3	0.27
4	0.19

Trend?

- The more you want to emphasize relevance to the **topic node set S** , the smaller you should set β .
 - A smaller β directs more votes $(1 - \beta)$ toward S in each iteration.
 - Drawback: The **general importance** of each page is also considered less

Example



$$S = \{1\}$$
$$\beta = 0.8$$

Node	Score
1	0.29
2	0.12
3	0.33
4	0.26

$$S = \{1,2\}$$
$$\beta = 0.8$$

Node	Score
1	0.26
2	0.20
3	0.29
4	0.23

$$S = \{1,2,3\}$$
$$\beta = 0.8$$

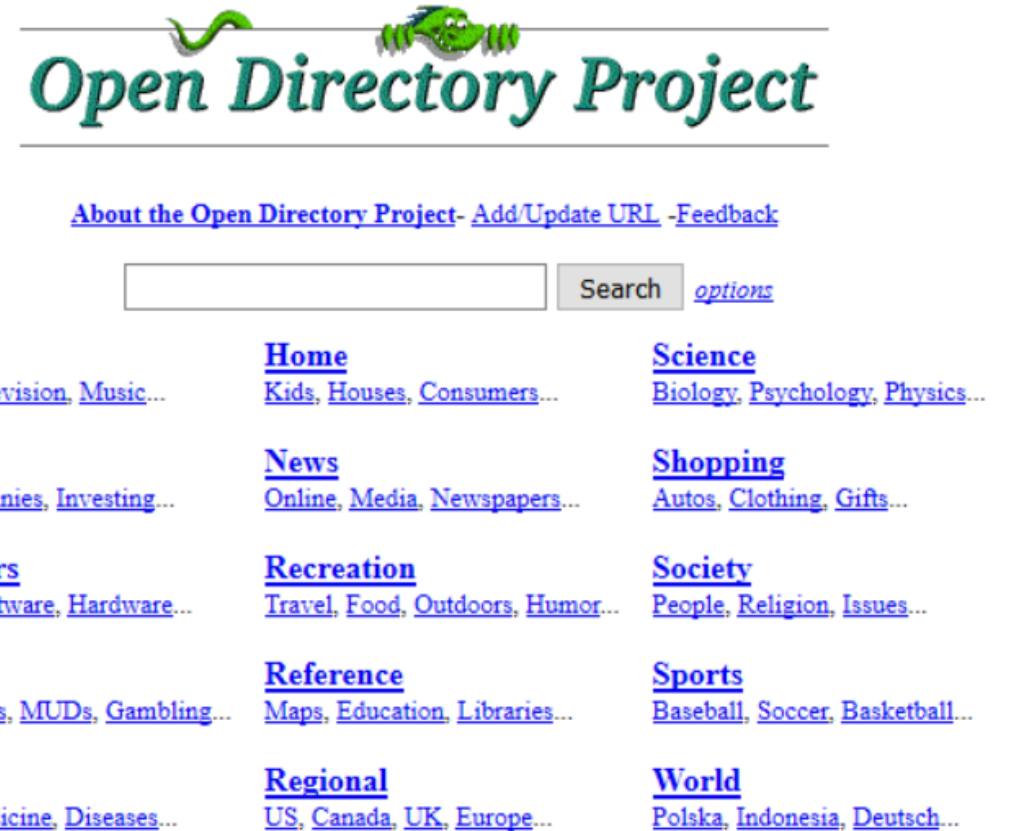
Node	Score
1	0.17
2	0.13
3	0.38
4	0.30

Trend?

- As S covers more nodes, relevance to the topic becomes increasingly less important.
- When S includes all nodes, **topic-sensitive PageRank** reduces to **standard PageRank**.

How to get S ?

- The 15 DMOZ top-level categories:
 - arts, business, sports, ...
 - Compute different PageRank scores for different topics
- Which topic ranking to use?
 - Users can pick from a menu
 - Classify the query into a topic
 - Query context, e.g., search history
 - User context, e.g., user's bookmarks



Questions?

Link Spamming

- Once Google became the dominant search engine, spammers began to work out ways to fool Google.
 - Imagine an “evil” user who, after creating his personal homepage, tries to manipulate its PageRank score to make it appear higher in people's search results.
- **Spam farms** were developed to concentrate PageRank on a single page.
- **Link spam**: Creating link structures that boost PageRank of a particular page



Link Spamming

- Three kinds of web pages from a spammer's point of view

- Inaccessible pages

- E.g., official homepage of CNN



- Accessible pages

- E.g., social media comment pages
 - The spammer can post links to his pages

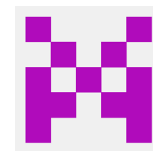
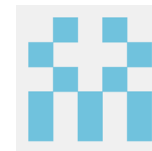


Black Friday **** Need copy and link****
6:00 AM - Nov 24, 2017
1,476 replies 22,851 retweets 72,463 likes

Reply: <https://XXX.github.io>

- Owned pages

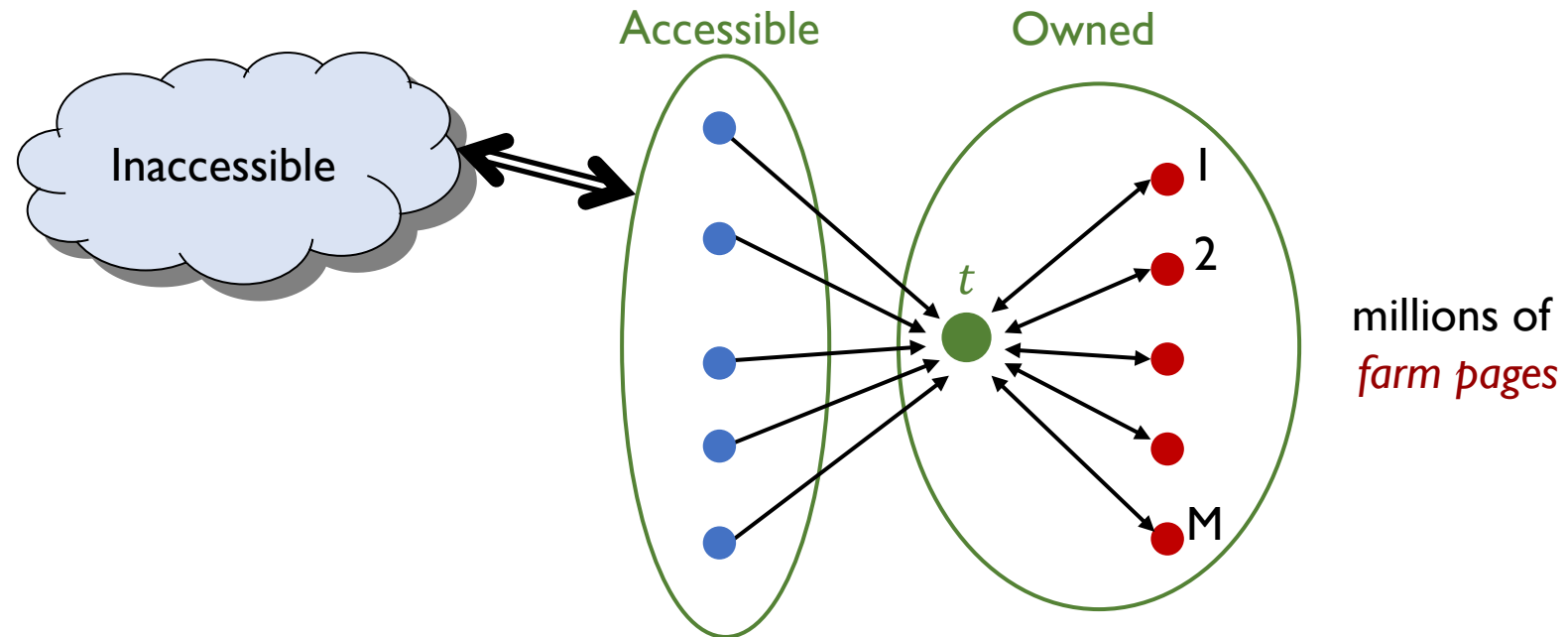
- Completely controlled by spammer
 - E.g., register several new GitHub accounts, and use each account to create a personal homepage.



...

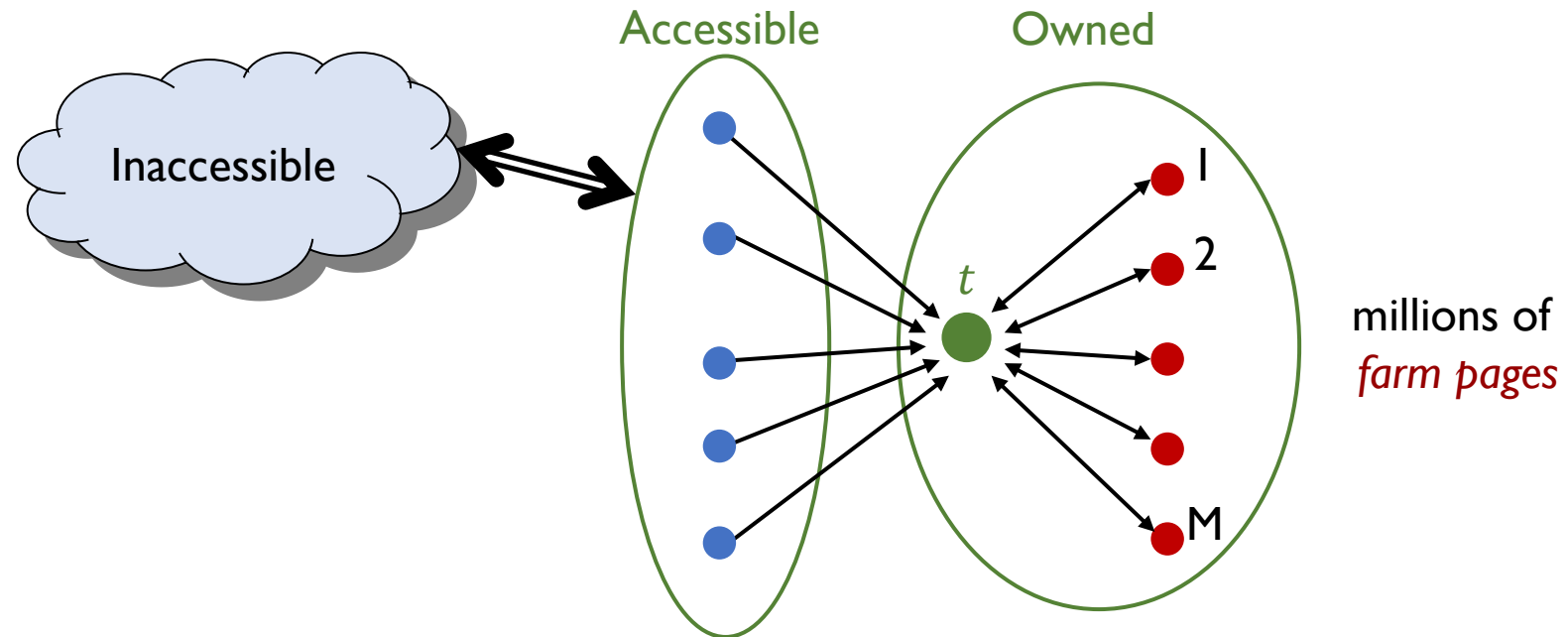
Link Farms

- **Spammer's goal:** Maximize the PageRank score of a target page t
- **Technique:**
 - Get as many links from accessible pages as possible to the target page t
 - Construct a “link farm” to get a PageRank multiplier effect



Analysis

- Let x be the PageRank score of the target page t
 - What is the PageRank score of each “farm” page? $\beta \frac{x}{M} + (1 - \beta) \frac{1}{N}$
- Let y be the PageRank scores contributed by accessible pages to t
- So $x = y + \beta M \left[\beta \frac{x}{M} + (1 - \beta) \frac{1}{N} \right] + (1 - \beta) \frac{1}{N}$

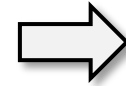


Analysis

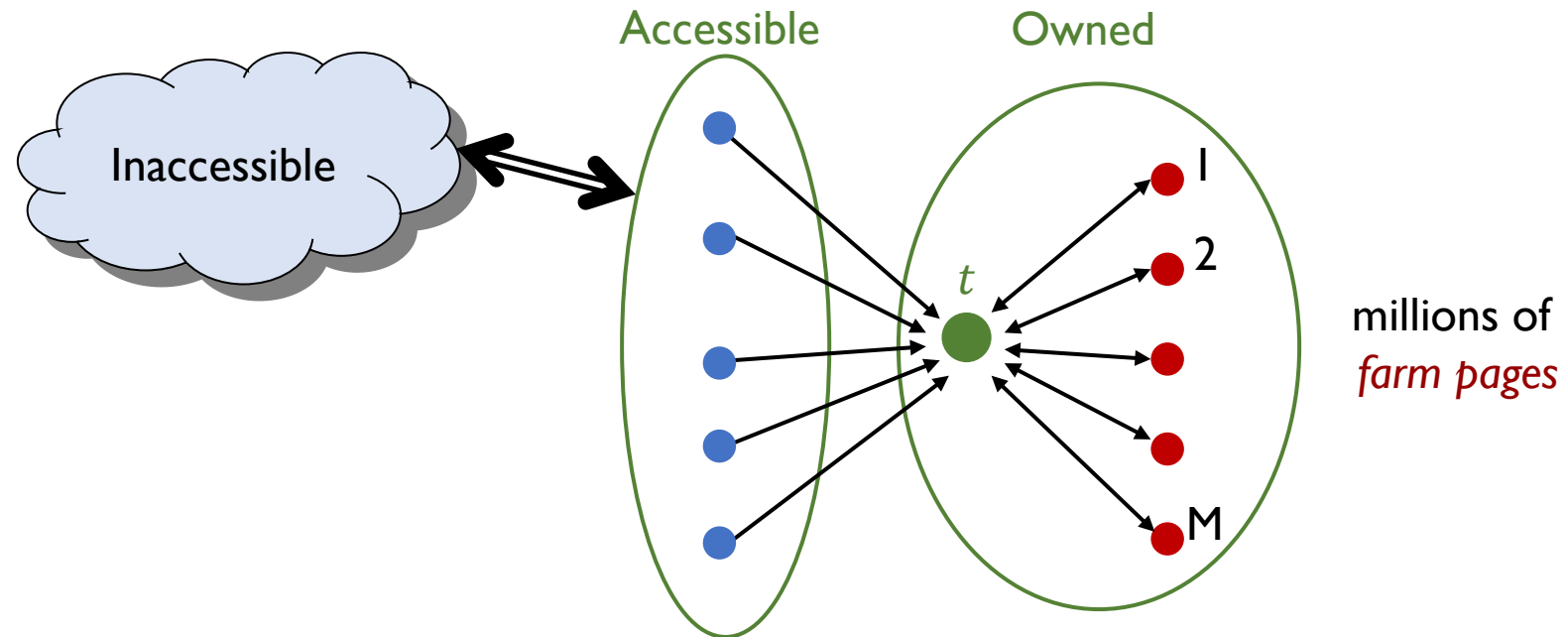
- Let x be the PageRank score of the target page t

- $$x = y + \beta M \left[\beta \frac{x}{M} + (1 - \beta) \frac{1}{N} \right] + (1 - \beta) \frac{1}{N}$$
$$= y + \beta^2 x + \frac{\beta(1-\beta)M}{N} + (1 - \beta) \frac{1}{N}$$

very small, can be ignored



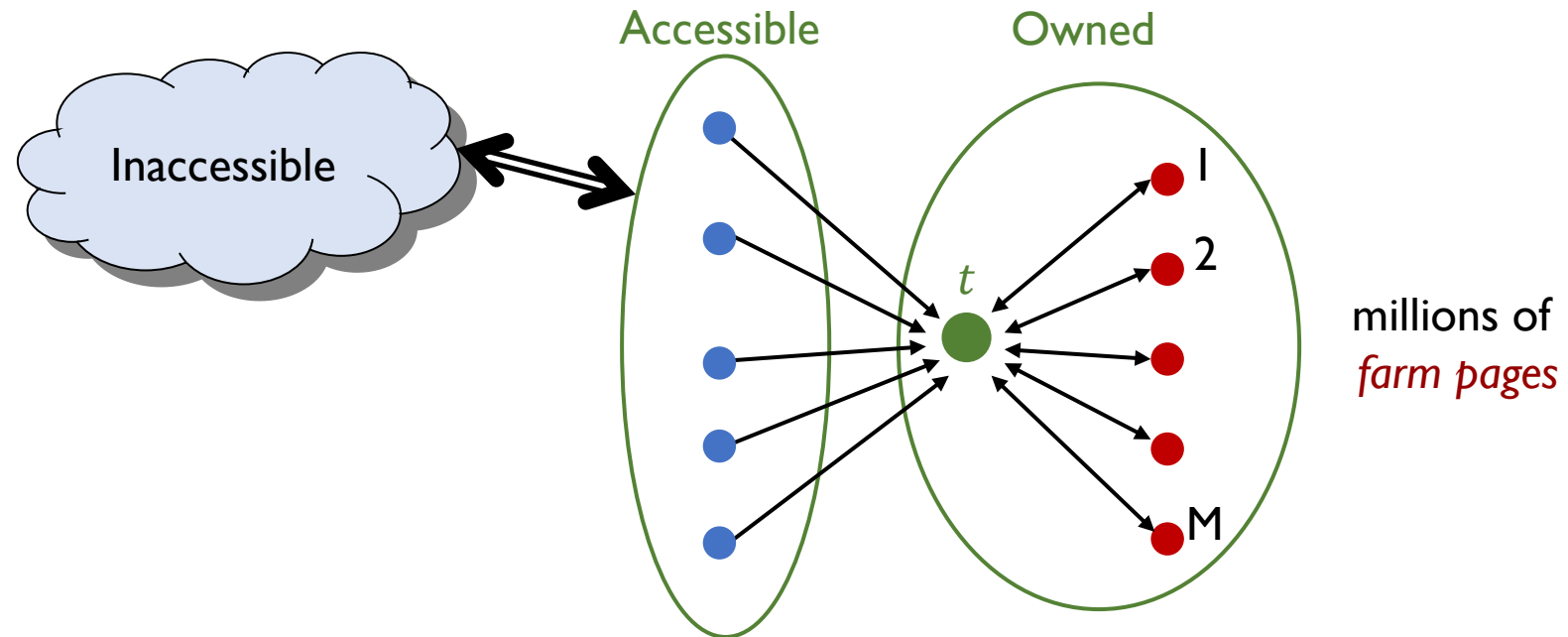
$$x = \frac{y}{1 - \beta^2} + \frac{\beta}{1 + \beta} \frac{M}{N}$$



Analysis

$$x = \frac{y}{1 - \beta^2} + \frac{\beta}{1 + \beta} \frac{M}{N}$$

- If $\beta = 0.8$, then $x = 2.78y + 0.44 \frac{M}{N}$
- By making M large, we can make x as large as we want



Extended Content
(will not appear in quizzes or the exam)

How to combat link spamming?

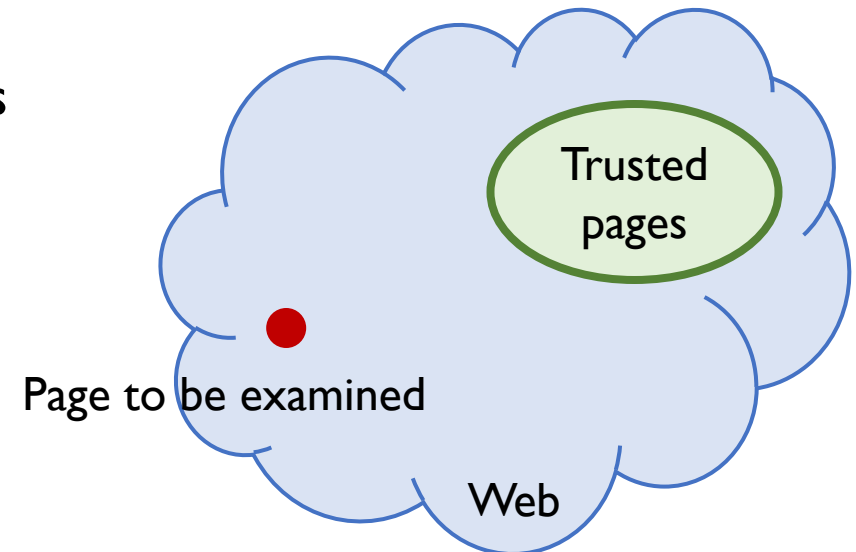
- **Naïve Idea:** detecting and blacklisting structures that look like spam farms
 - Leads to another war: hiding and detecting spam farms
- **More Advanced Idea:** **Topic-Sensitive PageRank** with teleportation to **trusted pages**
 - Example of **trusted pages**: *.edu* domains
- **Step 1:** Sample a set of seed pages from the web
 - Each page can be good (i.e., trusted) or bad (i.e., spam)
- **Step 2:** Ask humans to identify the good/bad pages in the seed set
 - An expensive task, so we must make seed set as small as possible

How to combat link spamming?

- **Step 1**: Sample a set of seed pages from the web
- **Step 2**: Ask humans to identify the good/bad pages in the seed set
- **Step 3**: Perform **Topic-Sensitive PageRank** with $S = \{\text{seed pages identified as good}\}$
 - Essentially propagate trust through links
 - Each page gets a trust value between 0 and 1
- Given a webpage, how to judge whether it is spam or not?
- **Solution 1**: Use a threshold value and mark all pages below the trust threshold as spam
 - Why should this work?
 - Are there cases where this may not work?

Spam Mass

- **Solution 1:** Use a threshold value and mark all pages below the trust threshold as spam
 - Are there cases where this may not work?
 - When will a node get a low **Topic-Sensitive PageRank** score?
 - **Case 1:** It is far away from S (i.e., trusted page)
 - **Case 2:** It has a low **Standard PageRank** score
 - This does not imply the node is a spam. Maybe it is just newly created.
- **Solution 2:** We can calculate what fraction of a page's PageRank comes from spam pages
 - In practice, we do not know all the spam pages, so we need to estimate.

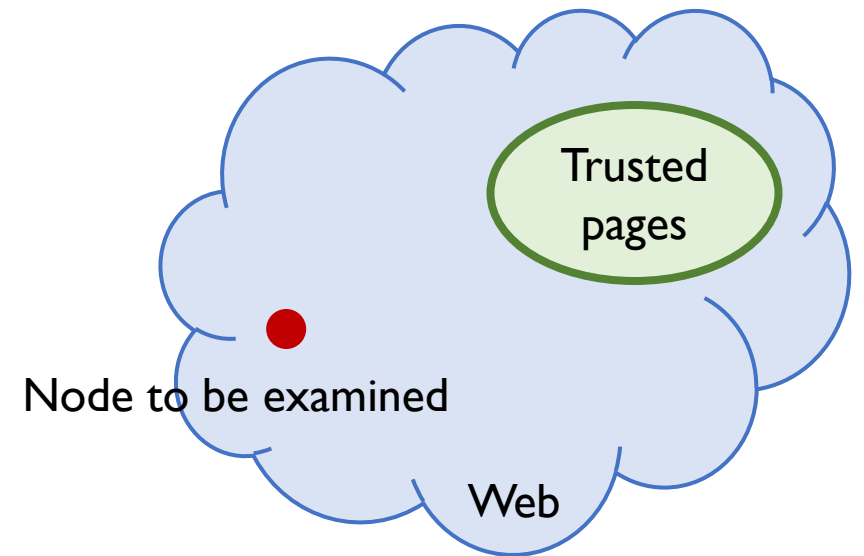


Spam Mass Estimation

- r_p = Standard PageRank score of page p
- r_p^+ = Topic-Sensitive PageRank of page p with teleportation into trusted pages only
 - r_p^+ may be small simply because r_p is small. We need to exclude this case.
- What **fraction** of a page's PageRank comes from spam pages?

$$r_p^- = r_p - r_p^+$$

- Spam mass of p is defined as $\frac{r_p^-}{r_p}$.
- Pages with high spam mass are judged as spam.





Thank You!

Course Website: <https://yuzhang-teaching.github.io/CSCE670-F26.html>